

Formula Collection

rendered with Lua $\LaTeX$

1. Quadratic Formula

Classic.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Pythagorean Theorem

Classic.

$$c = \sqrt{a^2 + b^2}$$

3. Sum of first n Squares

Classic.

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

4. Law of Cosines

Classic.

$$c^2 = a^2 + b^2 - 2ab \cos \angle C$$

5. Legendre's formula

Floors.

$$\nu_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$$

6. Euler's Identity

The most beautiful equation in mathematics.

$$e^{\pi i} + 1 = 0$$

7. Euler's Lesser-Known Identity

Troll.

$$\lceil e \rceil - \lfloor \pi \rfloor = 0$$

8. Normal Distribution

Thanks to Martin for correcting this!

$$\Phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

9. Fourier Transform

17 Equations That Changed the World.

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \omega} dx$$

10. Wave Equation

17 Equations That Changed the World.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

11. Navier-Stokes Equation

17 Equations That Changed the World.

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \mathbf{f}$$

12. Schrodinger's Equation

17 Equations That Changed the World.

$$i\hbar \frac{\partial}{\partial t} \Psi = H\Psi$$

13. Black-Scholes Equation

17 Equations That Changed the World.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

14. Relativity

17 Equations That Changed the World.

$$E = mc^2$$

15. Chaos Theory

17 Equations That Changed the World.

$$x_{t+1} = kx_t(1 - x_t)$$

16. Definition of the Derivative

17 Equations That Changed the World.

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

17. Euler's Formula for Polyhedra

17 Equations That Changed the World.

$$V - E + F = 2$$

18. Gravitation

17 Equations That Changed the World.

$$F = \frac{Gm_1m_2}{d^2}$$

19. AM-GM

Fun

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$

20. Stirling's Approximation

Fun

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

21. Stokes' Theorem

Fun

$$\iint_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r}$$

22. Divergence Theorem

Fun

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \oiint_S (\mathbf{F} \cdot \mathbf{n}) dS$$

23. Cauchy-Schwarz Inequality

Fun

$$|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \cdot \langle \mathbf{v}, \mathbf{v} \rangle$$

24. Area of a Circle

Simple

$$A = \pi r^2$$

25. Definition of τ

Troll.

$$\tau = 2\pi$$

26. Sophie Germain Identity

Simple.

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2)$$

27. Pascal's Identity

Classic.

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

28. Hockey-stick Identity

Classic.

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

29. Vandermonde's Identity

Classic.

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

30. Combinations

Simple.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

31. Heine's Identity

No idea what this is. Looks cool tho.

$$\frac{1}{\sqrt{z - \cos \psi}} = \frac{\sqrt{2}}{\pi} \sum_{m=-\infty}^{\infty} Q_{m-\frac{1}{2}}(z) e^{im\psi}$$

32. Binomial identity

Classic.

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

33. Hermite's Identity

Hadn't heard of this either.

$$\sum_{k=0}^{n-1} \left\lfloor x + \frac{k}{n} \right\rfloor = \lfloor nx \rfloor$$

34. Matrix Determinant Lemma

Or this lmao.

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u}) \det(\mathbf{A})$$

35. Euler Product of the Riemann-Zeta function

Classic.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-s}}$$

36. Irrationality of the Square Root of 2

I just really wanted to keep using `\|mathbb`.

$$\sqrt{2} \notin \mathbb{Q}$$

37. Heron's Formula

Classic.

$$[\triangle ABC] = \sqrt{s(s-a)(s-b)(s-c)}$$

38. Heisenberg's Uncertainty Principle

Classic.

$$\Delta x \Delta p \approx \hbar$$

39. Continued Fraction for $\pi/2$

@InertialObserver

$$\frac{\pi}{2} = 1 + \frac{1}{1 + \frac{1}{\frac{1}{2} + \frac{1}{\frac{1}{3} + \frac{1}{\frac{1}{4} + \dots}}}}$$

40. Sophomore's Dream

Cool.

$$\int_0^1 x^{-x} dx = \sum_{n=1}^{\infty} n^{-n}$$

41. Identity involving π and e

@InertialObserver

$$\prod_{n=2}^{\infty} e \left(1 - \frac{1}{n^2}\right)^{n^2} = \frac{\pi}{e\sqrt{e}}$$

42. Representation of the Golden Ratio

Classic

$$\phi = \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}}$$

43. The Sum of all Positive Integers

Troll.

$$\sum_{n=1}^{\infty} n = -\frac{1}{12}$$

44. Inverse of a complex number

Gotta know $\bar{\cdot}$ man

$$z^{-1} = \frac{\bar{z}}{|z|^2}, \forall z \neq 0$$

45. Definition of Convolution

Shout out to 6.003

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$

46. Definition of the Kronecker Delta function

{cases} ftw

$$\delta_{i,j} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

47. Bayes' Theorem

bae's theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

48. Probability Density Function of the Student's t -distribution

fun

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

49. De Morgan's laws

fun

$$\neg(P \wedge Q) \vdash (\neg P) \vee (\neg Q)$$

50. Principle of Inclusion-Exclusion

for dummies

$$|A \cup B| = |A| + |B| - |A \cap B|$$

51. General Principle of Inclusion-Exclusion

for galaxy brains

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \left| \bigcap_{j \in J} A_j \right|$$

52. Determinant of a 2×2 matrix

{matrix}

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

53. Sawtooth Function

mathbb cases floors, this has it all

$$S(x) = \begin{cases} x - \lfloor x \rfloor - 1/2 & x \in \mathbb{R} \setminus \mathbb{Z} \\ 0 & x \in \mathbb{Z} \end{cases}$$

54. Definition of Graham's Number

$G = g_{_ \{64\}}$

$$g_n = \begin{cases} 3 \uparrow \uparrow \uparrow \uparrow 3 & n = 1 \\ 3 \uparrow^{g_{n-1}} 3 & n \geq 2, n \in \mathbb{N} \end{cases}$$

55. Burnside's Lemma

The Lemma that is not Burnside's

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

56. Continuum Hypothesis

independent of ZFC!

$$\aleph_0 = |\mathbb{N}|, \mathfrak{c} = |\mathbb{R}| \nexists A : \aleph_0 < |A| < \mathfrak{c}$$

57. Spectral Decomposition

derived from memory

$$A = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix}^{-1}$$

58. Pythagorean Identity

basically just the Pythagorean theorem

$$\sin^2 \theta + \cos^2 \theta = 1$$

59. Double Angle for sin

back to basics

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$$

60. Double Angle for cos

back to basics

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

61. Fermat's Last Theorem

have a marvelous proof, but this description's too small to contain it

$$\nexists \{x, y, z, n\} \in \mathbb{N}, n > 2 : x^n + y^n = z^n$$

62. Fermat's Little Theorem

fermat's itty bitty theorem

$$a^p \equiv a \pmod{p}$$

63. Euler's Theorem

totients

$$\gcd(a, n) = 1 \implies a^{\varphi(n)} \equiv 1 \pmod{n}$$

64. QM-AM-GM-HM Inequality over 3 variables

cool-looking

$$\sqrt{\frac{a^2 + b^2 + c^2}{3}} \geq \frac{a + b + c}{3} \geq \sqrt[3]{abc} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

65. Extended Law of Sines

threw in the circumradius as well

$$\frac{a}{\sin \angle A} = \frac{b}{\sin \angle B} = \frac{c}{\sin \angle C} = 2R$$

66. Integration by Parts

it's just the product rule really

$$\int u dv = uv - \int v du$$

67. Definition of Perfect Numbers

shrug

$$\left\{ n : \sum_{d|n}^{d < n} d = n \right\}$$

68. Gaussian Integral

classic trick

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2 - y^2} dx dy} = \sqrt{\int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta} = \sqrt{\pi}$$

69. Definition of an Integral

why not

$$\int_a^b f(x)dx = \lim_{k \rightarrow \infty} \left((b-a) \sum_{i=1}^k \frac{f(a + i \frac{b-a}{k})}{k} \right)$$

70. Quantum Fourier transform

bra ket notation is fun

$$|x\rangle \mapsto \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_x^k |k\rangle$$

71. Recursive definition of the Hadamard transform

matrix in cases

$$H_m = \begin{cases} 1 & m = 0 \\ \frac{1}{\sqrt{2}} \begin{pmatrix} H_{m-1} & H_{m-1} \\ H_{m-1} & -H_{m-1} \end{pmatrix} & m > 0 \end{cases}$$

72. Wigner Transform of the Density Matrix

I know some of these words

$$W(x, p) = \frac{1}{\pi \hbar} \int_{-\infty}^{\infty} \langle x+y | \hat{\rho} | x-y \rangle e^{-2ipy/\hbar} dy$$

73. Imaginary numbers

Just gonna add some simple formulas

$$i^2 = -1$$

74. Sum of Cubes

Simple

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

75. RSA Decryption Algorithm

good ol' rivest

$$m = c^{e^{-1} \bmod \phi(n)} \pmod{n}$$

76. Contraposition

logic yo

$$(p \implies q) \iff (\neg q \implies \neg p)$$

77. Equation of a spring

Gonna use dots like the physicists do

$$m\ddot{x} = -kx$$

78. Sum of reciprocals of partial sums of $\mathbb{N}$

Credit to @IntertialObserver

$$\sum_{i=2}^{\infty} \frac{1}{\sum_{j=1}^i j} = 1$$

79. Binet's Formula

Classic

$$F_n = \frac{1}{\sqrt{5}} \left(\varphi^n - \frac{(-1)^n}{\varphi^n} \right)$$

80. Sum of first n Cubes

Classic

$$\sum_{k=0}^n k^3 = \left(\sum_{k=0}^n k \right)^2$$

81. The Basel Problem

Classic

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

82. Root Mean Square

how could i forget

$$f_{\text{rms}} = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} [f(t)]^2 dt}$$

83. The Harmonic Series

Classic

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

84. Tupper's Self-Referential Formula

Troll

$$\frac{1}{2} < \left\lfloor \text{mod} \left(\left\lfloor \frac{y}{17} \right\rfloor 2^{-17 \lfloor x \rfloor - \text{mod}(\lfloor y \rfloor, 17)}, 2 \right) \right\rfloor$$

85. Hölder's Inequality

Styled like the OTIS handouts by Evan Chen

$$\left(\sum_{i=1}^n a_i \right)^p \left(\sum_{i=1}^n b_i \right)^q \geq \left(\sum_{i=1}^n \sqrt[p+q]{a_i^p b_i^q} \right)^{p+q}$$

86. Rearrangement Inequality

kinda cool

$$a_1 \leq a_2 \leq \dots \leq a_n, b_1 \leq b_2 \leq \dots \leq b_n \implies \sum_{i=1}^n a_i b_i \geq \sum_{i=1}^n a_{\sigma(i)} b_i \geq \sum_{i=1}^n a_{n+1-i} b_i$$

87. Power Mean

like RMS-AM-GM-HM but like generalized

$$M_r(x_1, x_2, \dots, x_n) = \begin{cases} \left(\frac{1}{n} \sum_{i=1}^n x_i^r \right)^{1/r} & r \neq 0 \\ \sqrt[n]{\prod_{i=1}^n x_i} & r = 0 \end{cases}$$

88. Law of Tangents

yes this actually exists

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{\angle A - \angle B}{2}\right)}{\tan\left(\frac{\angle A + \angle B}{2}\right)}$$

89. Euler's Arctangent Identity

dammit euler OP

$$\tan^{-1}\left(\frac{1}{x}\right) = \tan^{-1}\left(\frac{1}{x+y}\right) + \tan^{-1}\left(\frac{y}{x^2 + xy + 1}\right)$$

90. The Dirichlet Convolution

bruh

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

91. Sum of a Row of Pascal's Triangle

not sure how else to word it

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

92. Alternating Harmonic Series

First use of $\ln$

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots = \ln 2$$

93. Definitions of Catalan's Constant

Credit to /u/heropup

$$G = \beta(2) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = \iint_{[0,1]^2} \frac{dx dy}{1+x^2 y^2}$$

94. Series Representation of Apéry's Constant

Credit to /u/heropup

$$\zeta(3) = \frac{5}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3 \binom{2n}{n}}$$

95. Definition of the Euler-Mascheroni Constant

Credit to /u/heropup

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \ln n \right) = \int_1^{\infty} \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx$$

96. Mertens' theorem

actually his third theorem

$$\prod_{p \in \mathbb{P}}^n \left(1 - \frac{1}{p} \right) \sim \frac{e^{-\gamma}}{\log n}$$

97. Green's First Identity

Credit to Varge

$$\int_{\Omega} (\psi \Delta \varphi + \nabla \psi \cdot \nabla \varphi) dV = \oint_{\partial \Omega} \psi (\nabla \varphi \cdot \mathbf{n}) dS$$

98. Cauchy-Riemann Equations

complex analysis is best analysis (1); credit to blu_bird

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

99. Cauchy's Integral Formula

complex analysis is best analysis (2); credit to blu_bird

$$f(z_0) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(z)}{z - z_0} dz$$

100. Cauchy's Differentiation Formula

complex analysis is best analysis (3); credit to blu_bird

$$f^{(k)}(z_0) = \frac{k!}{2\pi i} \oint_{\Gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz$$

101. Functional Equation for the Riemann-Zeta Function

This is the simplest example of a functional equation in the Langlands program. Conjecturally all Hasse-Weil zeta functions have Euler factorizations and functional equations with the Riemann zeta function as just one example.

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \pi^{-(1-s)/2} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s)$$

102. Well-ordering Principle

Classic. Credit to Eucruue

$$\forall M (M \subset \mathbb{N} \wedge M \neq \emptyset \implies \exists m_0 [m_0 \in M \wedge \forall n (n \in M \implies m \leq n)])$$

103. Asymptotic Formula for the Dirichlet Divisor Function

very cool dirichlet

$$\sum_{n \leq x} \tau(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x})$$

104. Prime Number Theorem

trivial

$$\pi(x) \sim \frac{x}{\log x}$$

105. Cumulative Distribution Function of the Gaussian Distribution

dense

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

106. Chernoff Bound

I never really learned what this was

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t}}$$

107. Union Bound

Never learned what this was either

$$\mathbb{P}\left(\bigcup_{i=1}^n X_i\right) \leq \sum_{i=1}^n \mathbb{P}(X_i)$$

108. Law of Total Probability

shrug

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)$$

109. Linear Least Squares Estimator

i love regression analysis

$$L[X|Y] = \mathbb{E}[X] + \frac{\text{cov}(X, Y)}{\text{var}(Y)}(Y - \mathbb{E}[Y])$$

110. Rademacher Complexity

The empirical Rademacher complexity of a function class

$$\mathcal{R}_n(\mathcal{F}) = \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(x_i) \right]$$

111. Definition of the Dilogarithm

aka Spence's function. don't wanna be accused of sleeping on spence

$$\text{Li}_2(z) = - \int_0^z \frac{\log(1-t)}{t} dt, z \in \mathbb{C}$$

112. Leibniz's Determinant Formula

Determinant of an n by n matrix

$$\det(A) = \sum_{\sigma \in S_n} \epsilon(\sigma) \prod_{i=1}^n A_{i, \sigma(i)}$$

113. Euler-Lagrange Equations

The basis for all of Lagrangian mechanics

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i}$$

114. Definition of the Euler Totient Function

what does totient mean anyways?

$$\varphi(n) = |\{k \in \mathbb{N}_{\leq n} \mid \gcd(k, n) = 1\}| = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

115. Sum of Divisors

i guess this person likes multiplicative functions

$$\sigma(n) = \sum_{d|n} d = \prod_{p^a || n} \left(\frac{p^{a+1} - 1}{p - 1} \right)$$

116. Einstein Field Equations

This form makes use of the Einstein tensor

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

117. Second Fundamental Theorem of Calculus

credit to VBG

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

118. Abel's Summation Formula

unclear to me why this is at all useful tbh

$$\sum_{x < n \leq y} a(n)f(n) = A(y)f(y) - A(x)f(x) - \int_x^y A(t)f'(t)dt$$

119. Lagrange's Theorem

more group theory

$$(G : H) = \frac{|G|}{|H|}$$

120. Catalan Numbers

A000108

$$C_n = \sum_{k=1}^{n-1} C_k C_{n-k-1} = \frac{1}{n+1} \binom{2n}{n}$$

121. Ising Model Hamiltonian

Mathematical model of ferromagnetism

$$H(\sigma) = - \sum_{\langle i,j \rangle} J_{ij} \sigma_i \sigma_j - \mu \sum_j h_j \sigma_j$$

122. Borwein Integral

The pattern famously breaks down after this integral.

$$\int_0^\infty \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} \dots \frac{\sin(x/13)}{x/13} dx = \frac{\pi}{2}$$

123. Wigner Semicircle Distribution

Essentially just a semicircle scaled to be a probability distribution.

$$f(x) = \begin{cases} \frac{2}{\pi R^2} \sqrt{R^2 - x^2} & -R \leq x \leq R \\ 0 & |x| > R \end{cases}$$

124. Parseval Gutzmer Formula

Apply the Cauchy Integral Formula to derive

$$f(z) = \sum_{k=0}^{\infty} a_k z^k \implies \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_{k=0}^{\infty} |a_k r^k|^2$$

125. Fubini's Theorem

switching the order of integration ftw

$$\int_X \left(\int_Y f(x, y) dy \right) dx = \int_Y \left(\int_X f(x, y) dx \right) dy = \int_{X \times Y} f(x, y) d(x, y)$$

126. Coarea Formula

A generalization of Fubini's theorem

$$\int_{\Omega} g(x) |\nabla u(x)| dx = \int_{\mathbb{R}} \left(\int_{u^{-1}(t)} g(x) dH_{n-1}(x) \right) dt$$

127. Equation of a Torus

yum, donuts

$$(\sqrt{x^2 + y^2} - R)^2 + z^2 = r$$

128. Ampère-Maxwell law

credit to Andrija

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

129. Gauss's Flux Theorem (differential form)

guess we're doing all of Maxwell's equations now huh

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

130. Gauss's law for Magnetism

I'll need to fix this once we discover magnetic monopoles.

$$\nabla \cdot \mathbf{B} = 0$$

131. Maxwell–Faraday equation

induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

132. Eigenvalue Formula

this yields the characteristic polynomial

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

133. Collatz Function

The conjecture is that repeated applications of this function always hit 1.

$$f(n) = \begin{cases} n/2 & n \equiv 0 \pmod{2} \\ 3n + 1 & n \equiv 1 \pmod{2} \end{cases}$$

134. Gamma Function

A generalization of the factorial function

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$$

135. Laplace Transform

signals and systems baby

$$\mathcal{L}\{f\}(s) = \int_0^\infty f(t) e^{-st} dt$$

136. Taylor Series

When $a = 0$, it's a Maclaurin series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

137. Quaternion Multiplication Formula

Hamilton famously carved this formula into the stone of a bridge when he came up with it.

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$$

138. General Solution to First-Order Linear Differential Equations

You can derive this with an integrating factor.

$$y = e^{-\int P(x)dx} \int Q(x)e^{\int P(x)dx} dx + Ce^{-\int P(x)dx}$$

139. Fibonacci Binomial Coefficients Identity

Sum up the shallow diagonals of Pascal's triangle to make Fibonacci numbers

$$F_{n+1} = \binom{n}{0} + \binom{n-1}{1} + \binom{n-2}{2} + \dots + \binom{n - \lfloor n/2 \rfloor}{\lfloor n/2 \rfloor}$$

140. Bellman Optimality Equation

Somehow connected to reinforcement learning! Credit to Constantine.

$$V^{\pi^*}(s) = \max_a \{R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^{\pi^*}(s')\}$$

141. Definition of a Well-founded Relation

R is well-founded iff every proper subset contains a minimal element with respect to R .
Credit to Constantine.

$$(\forall S \subseteq X)[S \neq \emptyset \implies (\exists m \in S)(\forall s \in S) \neg (sRm)]$$

142. Estimation Lemma

Credit to Ben Napier.

$$\left| \int_{\gamma} f(z) dz \right| \leq L(\gamma) \sup_{\gamma} |f|$$

143. Chaitin's Constant

The probability that a randomly constructed program will halt.

$$\Omega_F = \sum_{p \in P_F} 2^{-|p|}$$

144. Cauchy's Differentiation Formula

Credit to epm

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z-a)^{n+1}} dz$$

145. Defintion of the Quasi-Stationary Distribution

Getting rid of absorbing states.

$$\forall B \in \mathcal{B}(\mathcal{X}^a), \forall t \geq 0, P_{\nu}(Y_t \in B, T > t) = \nu(B)P_{\nu}(T > t)$$

146. Addition of Sound Levels in Decibels

50dB + 50dB -> ~53dB!

$$L_{ab} = 10 \log_{10} (10^{L_a/10} + 10^{L_b/10})$$

147. Fast-Growing Hierarchy

You wanna see some real speed?

$$f_{\alpha}(n) = \begin{cases} n+1 & \alpha = 0 \\ f_{\beta}(n) & \alpha = \beta + 1 \\ f_{\alpha[n]}(n) & \text{else} \end{cases}$$

148. Feigenbaum-Cvitanović Functional Equation

Damn, that's a mouthful.

$$g(g(x)) = -\frac{1}{\alpha}g(\alpha x)$$

149. Dirac Equation

Relativistic wave equation. Credit to Leon.

$$i\hbar\gamma^\mu\partial_\mu\psi - mc\psi = 0$$

150. Feynman's Trick

Essentially differentiating under the integral sign; the given problem is extremely difficult to solve otherwise. Credit to Aarsh Chotalia.

$$\int_0^\pi \ln(1 - 2\alpha \cos x + \alpha^2) dx = 2\pi \ln |\alpha|$$

151. Lorentz Factor

Time and length change by a factor of gamma when objects move near the speed of light.

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

152. Time Dilation

Clocks moving at high speed will be observed to tick slower.

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

153. Gauss's Flux Theorem (integral form)

Use the divergence theorem to get to the differential form.

$$\oiint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\epsilon_0}$$

154. Doppler Effect

beep beep beep

$$\frac{f_o}{f_s} = \frac{\lambda_s}{\lambda_o} = \frac{v \pm v_o}{v \mp v_s}$$

155. Bernoulli's Equation

I included just because it included this bonkers $\backslash\varrho$ thingy. What was wrong with $\backslash\rho$??

$$P_1 + \varrho g y_1 + \frac{1}{2} \varrho v_1^2 = P_2 + \varrho g y_2 + \frac{1}{2} \varrho v_2^2$$

156. Relation between K_p and K_c

Credit to Freddie Bullard.

$$K_p = K_c (RT)^{\Delta n}$$

157. Van der Waals Equation

Generalization of the Ideal Gas Law.

$$\left(P + a \frac{n^2}{V^2} \right) (V - nb) = nRT$$

158. Maxwell-Boltzmann Distribution

Don't have enough statistical mechanics formulas.

$$f(v) = 4\pi v^2 \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}}$$

159. Cayley-Hamilton Theorem

Square matrices over commutative rings are annihilated by their own characteristic polynomial.

$$p(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}) \implies p(\mathbf{A}) = 0$$

160. Chudnovsky's Formula for π

This formula, based on a Ramanujan formula, was used to calculate pi to the tens of trillions of digits.

$$\frac{1}{\pi} = 12 \sum_{k=0}^{\infty} \frac{(-1)^k (6k)! (545140134k + 13591409)}{(3k)! (k!)^3 (640320)^{3k+3/2}}$$

161. Residue Theorem

Q: Why did the mathematician name her dog Cauchy? A: Because it left a residue at every pole.

$$\frac{1}{2\pi i} \oint_{\gamma} f(z) dz = \sum_{p \text{ pole}} \mathbf{I}(\gamma, p) \text{Res}(f, p)$$

162. Center of Mass

In a uniform gravitation field, this is the same as the center of gravity.

$$\mathbf{R} = \frac{1}{M} \iiint_Q \rho(\mathbf{r}) \mathbf{r} dV$$

163. The Fundamental Group of the Circle

It's isomorphic to the group of integers. Credit to fish.

$$\pi_1(S^1) \cong \mathbb{Z}$$

164. Definition of the Operator Norm on a Finite Dimensional Banach Space.

Credit to Richik Chakraborty.

$$\left\{ \frac{\|T(x)\|'}{\|x\|} : x \neq 0, x \in X \right\} \equiv \{\|T(x)\|' : \|x\| = 1, x \in X\}$$

165. Green's Theorem

Credit to Facejo.

$$\oint_C (L dx + M dy) = \iint_D \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dx dy$$

166. Portfolio Variance

Used to compute the covariance of a portfolio made up of n different assets, if the single variances and covariances are known. Credit to Marco.

$$\sigma_z^2 = \left(\sum_{i=1}^n w_i^2 \sigma_i^2 \right) + 2 \left(\sum_{i=1}^{n-1} \sum_{j=i+1}^n w_i w_j \sigma_{i,j} \right)$$

167. Newton's Method

Credit to <https://github.com/lucasalavapena>.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

168. Shannon Entropy

Credit to <https://github.com/lucasalavapena>.

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

169. Pinsker's inequality

It's possible I'm off by a factor of two here.

$$\|\mu - \nu\|_{\text{TV}} \leq \sqrt{2D_{\text{KL}}(\mu\|\nu)}$$

170. Sackur-Tetrode equation

Entropy of monatomic ideal gas. Credit to Haydn Gwyn.

$$\frac{S}{k_B N} = \ln \left[\frac{V}{N} \left(\frac{4\pi m U}{3h^2 N} \right)^{3/2} \right] + \frac{5}{2}$$

171. Conditional Entropy

The amount of information needed to describe the outcome of a random variable given the outcome of another variable.

$$H(Y|X) = - \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)}$$

172. Force-Potential Relation

Force is defined as the negative gradient of the potential energy function. Credit to Mayank Kumar.

$$\mathbf{F} = -\frac{\partial U}{\partial x}\hat{\mathbf{i}} - \frac{\partial U}{\partial y}\hat{\mathbf{j}} - \frac{\partial U}{\partial z}\hat{\mathbf{k}} = -\vec{\nabla}(U)$$

173. Beta Function

A special function that is closely related to the gamma function and to binomial coefficients. Credit to Salil Gokhale.

$$B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$$

174. Moist Adiabatic Lapse Rate

The rate that the temperature falls with respect to altitude in a wet environment.

$$\Gamma_w = -\frac{dT}{dz} = g \frac{\left(1 + \frac{H_v r}{R_{sd} T}\right)}{\left(c_{pd} + \frac{H_v^2 r}{R_{sw} T^2}\right)}$$

175. Cardano's Formula

Solution for a depressed cubic. Credit to TetanicRain7592.

$$\sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$$

176. General Cubic Formula

The deltas represents the cubic's discriminants. You must choose /any/ cube root and /any/ square root that doesn't result in $C = 0$. Credit to TetanicRain7592.

$$C = \sqrt[3]{\frac{\Delta_1 \pm \sqrt{\Delta_1^2 - 4\Delta_0^3}}{2}}$$

177. Riemann Zeta Function

This formula works when the real part of s is greater than 1. Other cases require analytic continuation.

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$$

178. Tangent Sum of Angles Formula

Credit to TetanicRain7592.

$$\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 \mp \tan(\alpha) \tan(\beta)}$$

179. Inner Product of Continuous Complex Valued Functions

Credit to Zeus Hernández.

$$\langle f, g \rangle = \int_0^{2\pi} f(t) \overline{g(t)} dt$$

180. Definition of a Psuedorandom Generator

Crypto means Cryptography!

$$\left| \Pr_{x \leftarrow \{0,1\}^k} [\mathcal{A}(G(x)) = 1] - \Pr_{x \leftarrow \{0,1\}^{p(k)}} [\mathcal{A}(x) = 1] \right| < \mu(k)$$

181. Generalized Stokes' theorem

One theorem to rule them all

$$\int_{\partial M} \omega = \int_M d\omega$$

182. Cartan's magic formula

A cool little magic trick

$$\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$$

183. Ridge Regression

Estimate coefficients of a regression with L2 regularization

$$\ell(\mathbf{w}) = \frac{1}{N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

184. Evidence Lower Bound (ELBO)

Lower bound on the log-likelihood of observed data

$$L(\phi, \theta; x) = \mathbb{E}_{z \sim q_\phi(\cdot | x)} \left[\ln \frac{p_\theta(x, z)}{q_\phi(z | x)} \right]$$

185. Langevin Dynamics (Overdamped)

Stochastic gradient flow with a deterministic potential and random noise

$$dx_t = -\nabla U(x_t)dt + \sqrt{2\beta^{-1}}dW_t$$
